

Plant Physiology: The Architecture of Life

Plant physiology is the study of how plants function — from the molecular movements of water through cell membranes to the grand orchestration of seasonal flowering. Plants are not passive organisms; they are dynamic, self-regulating systems that integrate light, water, minerals, and chemical signals to grow, reproduce, and survive in ever-changing environments. This document explores the fundamental processes that define plant life, revealing the elegant interconnections between energy capture, nutrient use, and environmental response.

At the heart of plant physiology lies a continuous flow of energy and matter. Water absorbed from the soil travels through microscopic xylem vessels to reach leaves hundreds of feet above ground, driven by the physical forces of cohesion and transpiration. Minerals dissolved in that water are selectively absorbed and transported to fuel enzymatic reactions. Sunlight captured by chlorophyll is converted into chemical energy through photosynthesis, producing the sugars that power every other metabolic process in the plant. Respiration then breaks those sugars back down, releasing ATP — the universal energy currency — to drive biosynthesis, cell division, and active transport.

These core processes do not operate in isolation. They form a tightly integrated network where the output of one pathway becomes the input for another. Photosynthesis generates the carbohydrates that respiration consumes; respiration regenerates the ATP that powers mineral uptake and hormone signaling; hormones regulate the stomatal openings that control transpiration and CO₂ uptake; and environmental cues like day length and temperature synchronize developmental transitions through photoperiodism and vernalization. Understanding plant physiology means understanding this web of interdependencies — how a change in water availability can cascade through hormone levels, gene expression, and ultimately growth patterns.

From the microscopic scale of aquaporin channels and enzyme kinetics to the whole-plant scale of tropic movements and stress acclimation, every process described in these notes reflects millions of years of evolutionary refinement. Plants have solved problems of energy capture, resource allocation, and environmental sensing with a sophistication that continues to inspire modern biotechnology and agriculture. These notes synthesize the major topics of plant physiology into a coherent framework, emphasizing the critical integrations that make plant life possible.

PLANT BIOLOGY

PHYSIOLOGY

BIOCHEMISTRY

Water Relations and Osmosis: The Hydraulic Engine

Water is the lifeblood of plants, constituting up to 95% of the fresh weight of herbaceous tissues. The movement of water through plant systems is governed by **water potential (Ψ)**, a measure of the free energy status of water relative to pure water at standard conditions. Water always flows from regions of higher water potential to regions of lower water potential — a principle that underlies every water-related process in plant biology, from root absorption to leaf transpiration.

Osmosis is the diffusion of water across a selectively permeable membrane driven by differences in solute concentration. When a plant cell is placed in a solution with lower water potential than its cytoplasm, water enters the cell, increasing **turgor pressure** — the hydrostatic pressure exerted by the cell contents against the rigid cell wall. Turgor pressure is essential for maintaining cell rigidity, driving cell expansion during growth, and powering rapid movements such as stomatal opening and closing. Water crosses cell membranes through specialized protein channels called **aquaporins**, which dramatically increase membrane permeability to water and allow rapid equilibration of water potential.

Water Potential Components

$\Psi = \Psi_s$ (solute potential) + Ψ_p (pressure potential) + Ψ_m (matric potential). Solutes lower Ψ_s ; turgor raises Ψ_p .

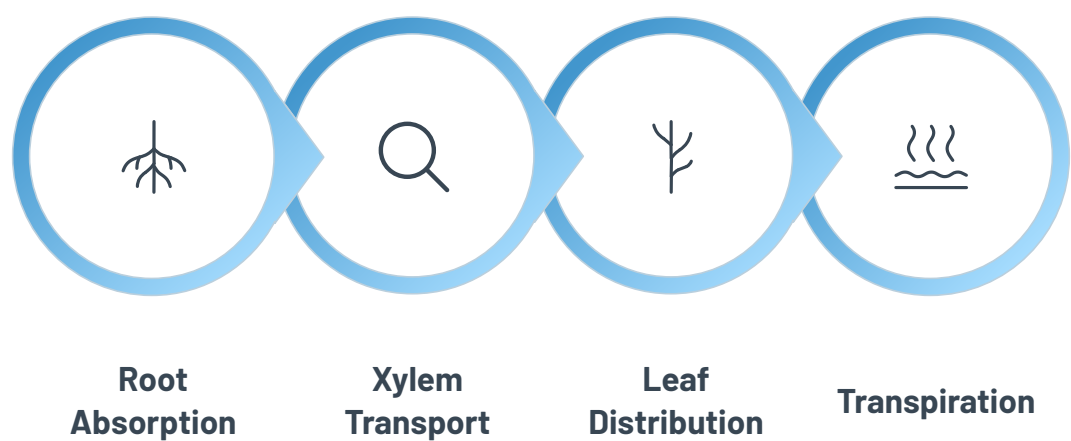
Osmosis & Turgor

Water enters cells via aquaporins until Ψ equilibrium is reached, generating turgor pressure critical for cell expansion and structural support.

Cohesion-Tension Theory

Transpiration creates negative pressure (tension) in xylem; water's cohesive and adhesive properties allow continuous columns to be pulled upward without breaking.

The ascent of water through the xylem is explained by the **Cohesion-Tension Theory**. As water evaporates from the mesophyll cell walls into the substomatal cavity and exits through open stomata (transpiration), it creates a negative pressure or tension in the xylem vessels. Because water molecules are highly cohesive — held together by hydrogen bonds — this tension is transmitted all the way down through the continuous water column to the roots, effectively pulling water upward against gravity. The adhesive forces between water and the hydrophilic xylem walls further stabilize this column. Transpiration rates are influenced by environmental factors including humidity, temperature, wind, and light intensity, all of which affect the vapor pressure gradient between the leaf interior and the surrounding air.



This diagram illustrates the continuous pathway of water through the plant, from soil absorption through the roots to evaporation from leaf surfaces — a process driven entirely by water potential gradients and the physical properties of water itself.

Mineral Nutrition: Fueling Growth

Plants require a diverse array of mineral elements to sustain their metabolic activities, structural integrity, and reproductive success. These **essential elements** are classified into macronutrients — required in large quantities — and micronutrients — needed in trace amounts but equally critical for enzyme function. The primary macronutrients include nitrogen (N), phosphorus (P), and potassium (K), while secondary macronutrients include calcium (Ca), magnesium (Mg), and sulfur (S). Micronutrients such as iron (Fe), manganese (Mn), zinc (Zn), copper (Cu), boron (B), molybdenum (Mo), and chlorine (Cl) serve as cofactors for numerous enzymatic reactions.

Mineral absorption occurs primarily through root hairs and the epidermal cells of young roots. Ions enter root cells via two main mechanisms: **passive transport** (diffusion through ion channels down electrochemical gradients) and **active transport** (energy-dependent movement against gradients). The **Proton ATPase (H⁺-ATPase)** pump located in the plasma membrane is central to active mineral uptake. By pumping protons out of the cell, it creates an electrochemical gradient — both a pH gradient and a membrane potential — that drives the secondary active transport of ions such as nitrate (NO₃⁻), potassium (K⁺), and phosphate (H₂PO₄⁻) through symporters and antiporters.

Once inside the root, minerals travel to the xylem either through the **apoplastic pathway** (through cell walls and intercellular spaces) or the **symplastic pathway** (through plasmodesmata connecting cytoplasm). The Casparian strip in the endodermis acts as a selective barrier, forcing all ions to pass through cell membranes and allowing the plant to regulate which minerals enter the vascular system. Deficiency symptoms appear when essential nutrients are unavailable in sufficient quantities. Nitrogen deficiency causes chlorosis (yellowing) of older leaves; phosphorus deficiency leads to purple discoloration and stunted growth; potassium deficiency produces marginal leaf necrosis. Each deficiency pattern reflects the specific metabolic role of the missing element.

Macronutrients

- Nitrogen (N) — proteins, nucleic acids
- Phosphorus (P) — ATP, membranes
- Potassium (K) — osmoregulation
- Calcium (Ca) — cell walls, signaling
- Magnesium (Mg) — chlorophyll center
- Sulfur (S) — amino acids

Micronutrients

- Iron (Fe) — electron transport
- Zinc (Zn) — enzyme cofactor
- Manganese (Mn) — water splitting
- Copper (Cu) — redox reactions
- Boron (B) — cell wall formation
- Molybdenum (Mo) — nitrogen fixation

17

Essential Elements

Total number of mineral elements required for complete plant life cycles

9

Macronutrients

Elements needed in quantities greater than 0.1% dry weight

8

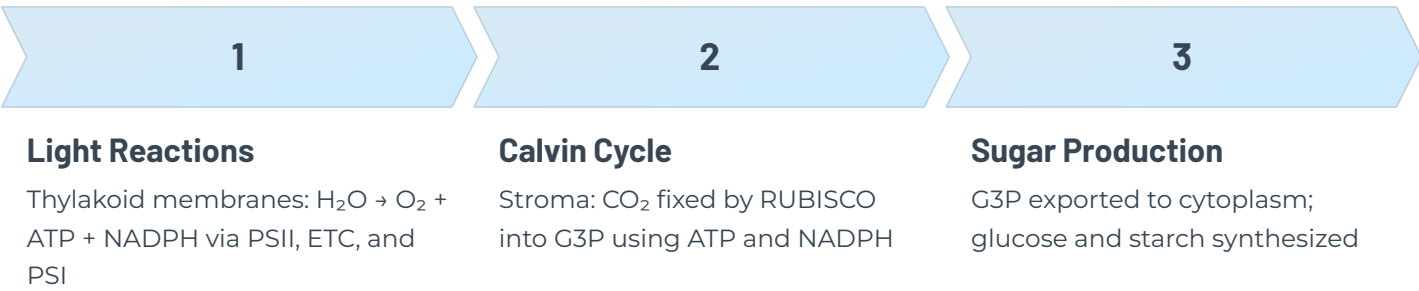
Micronutrients

Trace elements required in minute but critical concentrations

Photosynthesis: Capturing the Sun

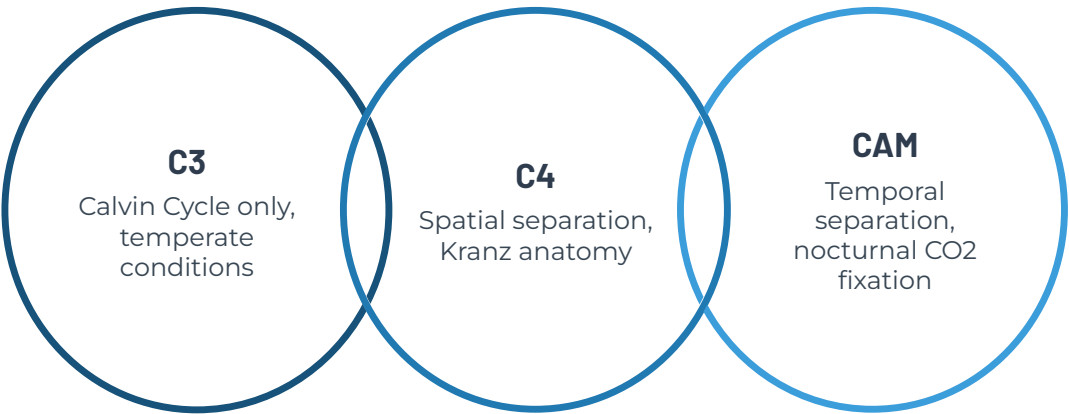
Photosynthesis is the foundational process by which plants convert light energy into chemical energy, producing the organic compounds that fuel virtually all life on Earth. It occurs in two interconnected phases: the **light-dependent reactions** and the **Calvin Cycle (light-independent reactions)**. The light reactions take place in the thylakoid membranes of chloroplasts, where pigment molecules — primarily chlorophyll a, chlorophyll b, and carotenoids — absorb photons and use their energy to drive electron transport chains.

During the light reactions, water molecules are split at **Photosystem II (PSII)** in a process called photolysis, releasing oxygen as a byproduct, protons that contribute to the proton gradient, and electrons that replace those lost by chlorophyll. These electrons travel through an electron transport chain involving plastoquinone, the cytochrome b₆f complex, and plastocyanin, ultimately reaching **Photosystem I (PSI)**. The energy released during electron transport is used to pump protons into the thylakoid lumen, creating a proton motive force that drives ATP synthesis via **ATP synthase** (photophosphorylation). At PSI, electrons are re-energized by light and transferred to NADP⁺ via ferredoxin, producing **NADPH**. Both ATP and NADPH are then used in the Calvin Cycle.



The **Calvin Cycle** operates in the chloroplast stroma and uses the ATP and NADPH from the light reactions to fix atmospheric CO_2 into organic molecules. The key enzyme **RUBISCO (Ribulose-1,5-bisphosphate carboxylase/oxygenase)** catalyzes the carboxylation of ribulose-1,5-bisphosphate (RuBP), producing two molecules of 3-phosphoglycerate (3-PGA). After reduction and regeneration phases, the cycle produces glyceraldehyde-3-phosphate (G3P), which can be used to synthesize glucose, starch, or other organic compounds.

Plants have evolved alternative carbon fixation pathways to cope with environmental stress. **C4 plants** (e.g., maize, sugarcane) spatially separate initial CO_2 fixation (in mesophyll cells via PEP carboxylase) from the Calvin Cycle (in bundle sheath cells), concentrating CO_2 around RUBISCO and minimizing photorespiration. **CAM plants** (e.g., cacti, pineapple) temporally separate these processes, fixing CO_2 at night and running the Calvin Cycle during the day — a crucial adaptation for water-limited environments. Both pathways represent evolutionary solutions to the fundamental problem of photorespiration, where RUBISCO mistakenly fixes O_2 instead of CO_2 under hot, dry conditions.



This diagram compares the three major photosynthetic pathways, highlighting how C4 and CAM plants have evolved structural and temporal adaptations to minimize photorespiration and thrive in challenging environments.

Respiration and Metabolism: The Energy Currency

Cellular Respiration Pathway

Cellular respiration is the process by which plants break down organic molecules — primarily glucose produced during photosynthesis — to regenerate **ATP**, the universal energy currency of the cell. Respiration occurs in three major stages: **glycolysis** (in the cytoplasm), the **tricarboxylic acid (TCA) cycle** or Krebs cycle (in the mitochondrial matrix), and **oxidative phosphorylation** (at the inner mitochondrial membrane).

During glycolysis, one molecule of glucose (6C) is split into two molecules of pyruvate (3C), producing a net gain of 2 ATP and 2 NADH. Pyruvate then enters the mitochondria, where it is decarboxylated to acetyl-CoA and fed into the TCA cycle. The TCA cycle completes the oxidation of carbon compounds, releasing CO₂ and generating NADH, FADH₂, and GTP/ATP. These reduced electron carriers (NADH and FADH₂) donate electrons to the mitochondrial electron transport chain, where the energy released drives proton pumping and ultimately ATP synthesis via ATP synthase — a process yielding approximately 30–32 ATP per glucose molecule under aerobic conditions.

The intimate connection between photosynthesis and respiration forms the core metabolic cycle of plant life. Photosynthesis stores energy in the chemical bonds of glucose; respiration releases that energy in usable form as ATP. The carbon skeletons generated during respiration also serve as precursors for the biosynthesis of amino acids, lipids, nucleotides, and secondary metabolites. **Metabolic flux** — the rate at which metabolites flow through these pathways — is tightly regulated by cellular energy charge (the ratio of ATP:ADP:AMP), allosteric enzyme regulation, and hormonal signals that coordinate growth with energy availability.

Energy Flow Integration

Photosynthesis

Light → ATP + NADPH
→ Glucose (energy storage)

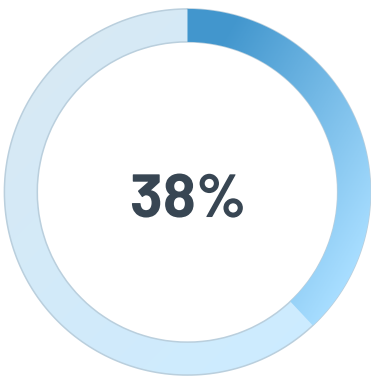
Respiration

Glucose → ATP + CO₂ + H₂O (energy release)

ATP

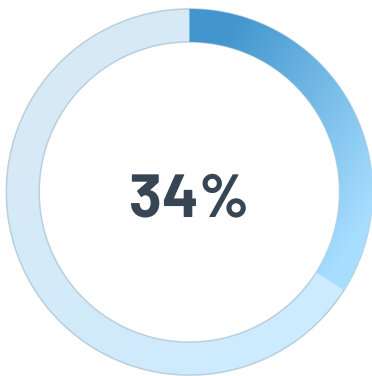
Universal currency linking all metabolic processes

Key Integration: The ATP produced by respiration powers the active transport of minerals, the biosynthesis of macromolecules, cell division, and the maintenance of ion gradients — essentially every energy-requiring process in the plant.



Energy as Heat

Approximate percentage of glucose energy released as heat during respiration



ATP Efficiency

Estimated efficiency of ATP capture from glucose oxidation under optimal conditions



Carbon Cycling

All carbon fixed in photosynthesis is eventually respired or incorporated into biomass

Nitrogen Metabolism: The Building Blocks

Nitrogen is a fundamental component of amino acids, proteins, nucleic acids, chlorophyll, and many coenzymes — making nitrogen metabolism one of the most critical processes in plant biology. Plants acquire nitrogen primarily in inorganic forms: **nitrate (NO_3^-)** and **ammonium (NH_4^+)**, absorbed from the soil through root transporters. Some plants, particularly legumes, also benefit from **biological nitrogen fixation** through symbiotic relationships with rhizobia bacteria in root nodules, which convert atmospheric N_2 into ammonia using the enzyme nitrogenase.

Once absorbed, nitrate must be reduced before it can be incorporated into organic molecules. This occurs in two enzymatic steps: **nitrate reductase** (in the cytoplasm) reduces NO_3^- to NO_2^- , and **nitrite reductase** (in chloroplasts or plastids) further reduces NO_2^- to NH_4^+ . Ammonium is then assimilated into amino acids primarily through the **GS-GOGAT cycle** (Glutamine Synthetase–Glutamate Synthase pathway). Glutamine synthetase incorporates NH_4^+ into glutamate to form glutamine, and glutamate synthase transfers the amide nitrogen to 2-oxoglutarate, regenerating glutamate. From glutamate and glutamine, nitrogen is distributed to all other amino acids through transamination reactions.

01

Nitrogen Uptake

NO_3^- and NH_4^+ absorbed from soil via specific root transporters; energy-dependent active transport

02

Nitrate Reduction

$\text{NO}_3^- \rightarrow \text{NO}_2^-$ (nitrate reductase) $\rightarrow \text{NH}_4^+$ (nitrite reductase); requires NADH/NADPH and ferredoxin

03

GS-GOGAT Cycle

NH_4^+ assimilated into glutamine and glutamate; the central gateway to all amino acid synthesis

04

Amino Acid Synthesis

Transamination distributes nitrogen to all 20 amino acids; linked to carbon skeletons from TCA cycle

Nitrogen metabolism is extraordinarily energy-intensive. The reduction of nitrate to ammonium and the subsequent GS-GOGAT cycle consume significant amounts of ATP and reducing power (NADPH), directly coupling nitrogen assimilation to the carbohydrate metabolism generated by photosynthesis. This coupling ensures that nitrogen assimilation is coordinated with the plant's energy status — when photosynthesis is active and carbohydrates are abundant, nitrogen assimilation accelerates to support protein synthesis and growth. Conversely, under low-light or stress conditions, nitrogen metabolism is downregulated to conserve energy. The enzyme **nitrate reductase** itself is subject to complex regulation by light, carbohydrates, and hormonal signals, serving as a key integration point between carbon and nitrogen metabolism.

Plant Hormones: Chemical Signaling

Plant hormones (phytohormones) are low-molecular-weight organic compounds that act as chemical messengers, coordinating growth, development, and stress responses across the entire plant. Unlike animal hormones, plant hormones are not produced in specialized glands but are synthesized in many tissues and can act locally or be transported to distant target cells. They function through complex **signal transduction pathways**, often involving second messengers such as calcium ions (Ca^{2+}), inositol trisphosphate (IP_3), and reactive oxygen species (ROS), as well as protein phosphorylation cascades mediated by kinases and phosphatases.



Auxins (IAA)

Promote cell elongation, apical dominance, and root initiation. Act through the TIR1/AFB receptor pathway, triggering degradation of Aux/IAA repressor proteins and activating ARF transcription factors. Critical for phototropism and gravitropism.



Gibberellins (GA)

Stimulate stem elongation, seed germination, and flowering. Break seed dormancy by inducing α -amylase synthesis in cereal aleurone layers. The DELLA protein represses GA responses; GA triggers DELLA degradation via the ubiquitin-proteasome pathway.



Cytokinins

Promote cell division, delay senescence, and stimulate shoot formation in tissue culture. Act through a two-component histidine kinase signaling system. The auxin:cytokinin ratio determines root vs. shoot differentiation in callus tissue.



Ethylene

A gaseous hormone regulating fruit ripening, leaf abscission, and stress responses. Biosynthesized from methionine via ACC. Perceived by membrane-bound receptors that act as negative regulators; ethylene binding inactivates the repressors, activating downstream responses.



Absciscic Acid (ABA)

The primary stress hormone; induces stomatal closure during drought, promotes seed dormancy, and regulates stress-responsive gene expression. Acts through PYR/PYL receptors and PP2C phosphatases, ultimately activating SnRK2 kinases that phosphorylate target proteins.

Signal Integration: Hormones rarely act in isolation. The balance and interaction between hormones — such as the antagonism between ABA (dormancy) and GA (germination), or between auxin (apical dominance) and cytokinin (lateral bud growth) — determines the ultimate physiological outcome. Many hormone pathways converge on **Calcium-Calmodulin** cascades and MAP kinase signaling modules, providing a unified framework for signal integration.

Photoperiodism and Vernalization: The Biological Calendar

Photoperiodism

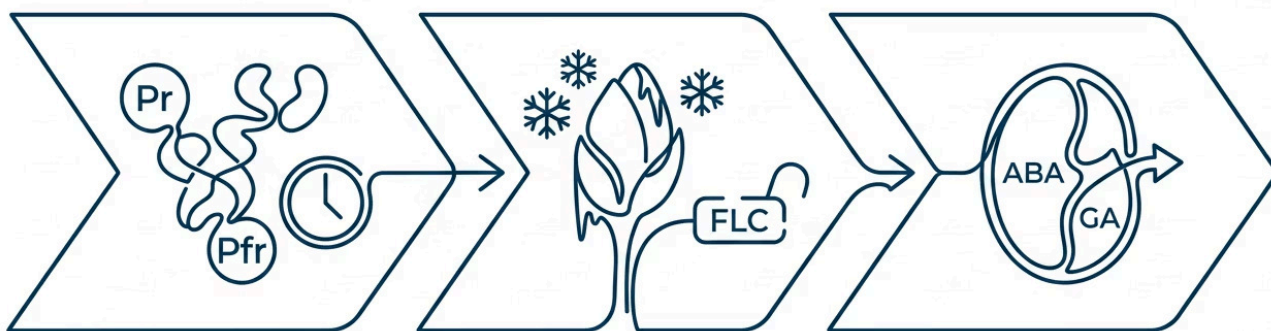
Photoperiodism is the physiological response of plants to the relative lengths of light and dark periods, enabling them to synchronize critical developmental events — particularly flowering — with favorable seasons. Plants are classified as **short-day plants** (flower when nights exceed a critical length, e.g., chrysanthemum, rice), **long-day plants** (flower when nights are shorter than critical, e.g., spinach, wheat), or **day-neutral plants** (flowering independent of day length, e.g., tomato).

The primary photoreceptor mediating photoperiodic responses is **phytochrome**, a protein pigment existing in two interconvertible forms: **Pr** (red-light absorbing, biologically inactive) and **Pfr** (far-red-light absorbing, biologically active). Red light (660 nm) converts Pr to Pfr; far-red light (730 nm) reverses this. During the day, sunlight enriches Pfr; during the night, Pfr slowly reverts to Pr or is degraded. The ratio of Pfr:Pr at dawn provides the plant with a precise measure of night length. In long-day plants, sufficient Pfr at the end of the night promotes flowering through the **CONSTANS (CO) → FLOWERING LOCUS T (FT)** pathway; in short-day plants, the opposite occurs.

Vernalization & Seed Dormancy

Vernalization is the acquisition of a plant's ability to flower after prolonged exposure to cold temperatures. This mechanism ensures that flowering occurs in spring rather than autumn, preventing reproductive failure due to frost. At the molecular level, vernalization silences the **FLC (FLOWERING LOCUS C)** gene — a repressor of flowering — through epigenetic modifications (histone methylation) that are mitotically stable but reset in the next generation. This allows winter cereals like wheat and barley to flower only after experiencing winter cold.

Seed dormancy is an adaptive strategy that prevents germination under unfavorable conditions. Dormancy is controlled by the balance between **ABA** (which maintains dormancy) and **GA** (which breaks it). Environmental cues such as cold stratification, light exposure (via phytochrome), nitrate availability, and after-ripening time can break dormancy by shifting this hormonal balance. This ensures that seeds germinate only when conditions are suitable for seedling survival.



PHOTOPERIODISM.
Day length detection
by phytochrome Pr/Pfr.
Regulates CO/FT
flowering pathway.

VERNALIZATION.
Cold exposure required.
Silences FLC repressor
via epigenetics, ensuring
spring blooms.

SEED DORMANCY.
Controlled by ABA vs
GA hormone balance.
Broken by cold, light,
or after-ripening.

These three mechanisms — photoperiodism, vernalization, and seed dormancy — work in concert to ensure that plants complete their life cycles at optimal times, integrating light quality, temperature history, and hormonal status into a unified seasonal calendar.

Plant Movements and Stress Physiology

Although rooted in place, plants are far from immobile. They exhibit a remarkable array of movements in response to environmental stimuli, and they possess sophisticated defense systems to cope with biotic and abiotic stresses. These capabilities reflect the evolutionary imperative to optimize resource capture, avoid damage, and survive in dynamic environments.

Tropic Movements

Directional growth responses toward or away from a stimulus. **Phototropism** (toward light) is mediated by auxin redistribution via PIN proteins, causing differential cell elongation on the shaded side. **Gravitropism** (response to gravity) involves statoliths (amyloplasts) settling in root cap cells, triggering asymmetric auxin distribution. **Thigmotropism** (response to touch) guides climbing plants around supports.

Nastic Movements

Non-directional movements independent of stimulus direction, driven by rapid changes in **turgor pressure**. The classic example is the rapid leaf folding of *Mimosa pudica* in response to touch, mediated by ion fluxes (K^+ and Cl^-) in pulvinar motor cells that cause water efflux and cell collapse. Stomatal movements are another example, controlled by guard cell turgor regulated by ABA, light, and CO_2 .

Abiotic Stress Responses

Plants face salinity, drought, heat, cold, and UV radiation. Responses include: osmotic adjustment via compatible solutes (proline, glycine betaine); antioxidant enzyme induction (SOD, catalase, ascorbate peroxidase) to scavenge ROS; heat shock protein (HSP) synthesis for protein protection; and membrane lipid remodeling to maintain fluidity. **ABA** is the master regulator of drought responses, triggering stomatal closure and stress gene expression.

Biotic Stress & Defense Signaling

Pathogen attack activates two major defense hormone pathways: **Salicylic Acid (SA)** mediates resistance against biotrophic pathogens through systemic acquired resistance (SAR) and PR protein induction; **Jasmonic Acid (JA)** and its derivative **Methyl Jasmonate** defend against necrotrophs and herbivores, often inducing protease inhibitors and secondary metabolites. These pathways can antagonize each other, allowing plants to prioritize defenses based on the threat type.



Heat Stress

HSP synthesis, membrane stabilization, ROS scavenging



Drought

ABA-mediated stomatal closure, osmolyte accumulation



Salinity

Ion compartmentalization, Na^+/H^+ antiporters, compatible solutes

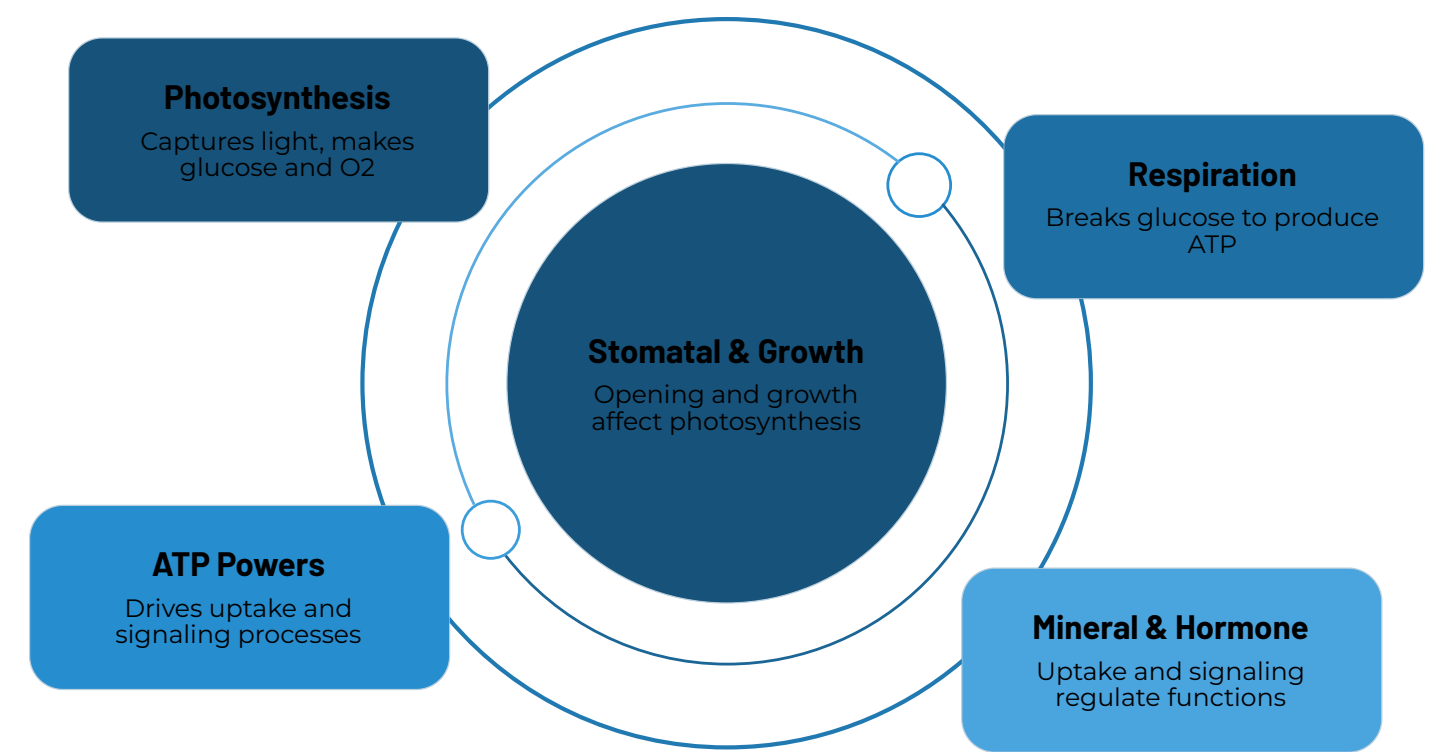


Pathogen Defense

SA/JA signaling, PR proteins, hypersensitive response

Conclusion: The Integrated Plant

Plant physiology reveals a remarkable truth: every process in a plant is connected to every other. Water absorbed through roots fuels the turgor that drives cell expansion; minerals transported in that water serve as enzyme cofactors for the reactions that fix carbon; carbon fixed by photosynthesis provides the substrate for respiration; respiration generates the ATP that powers mineral uptake, hormone synthesis, and stress defense; and hormones coordinate the timing and intensity of all these processes in response to environmental cues. This is not a collection of isolated systems — it is a single, integrated metabolic network.



This cycle illustrates the fundamental integration of plant physiological processes — each step both depends on and influences the others, creating a self-regulating system that continuously adapts to internal and external conditions.

The Central Metabolic Integration

The most critical integration in plant physiology is the **photosynthesis–respiration–ATP–metabolism** axis. Photosynthesis captures solar energy and stores it in glucose; respiration releases that energy as ATP; ATP drives every energy-requiring process including active mineral transport, biosynthesis of proteins and nucleic acids, cell division, stomatal movements, and the synthesis of hormones and secondary metabolites. When stress occurs, this axis is dynamically adjusted: ABA closes stomata to conserve water (reducing CO₂ uptake and photosynthesis), respiration may increase to provide energy for stress responses, and nitrogen metabolism is redirected toward stress-protective compounds like proline and pathogenesis-related proteins.

Understanding plant physiology at this integrative level is essential for addressing global challenges in agriculture, climate change, and food security. Crop improvement strategies — whether through conventional breeding or biotechnology — depend on manipulating these interconnected pathways to enhance yield, stress tolerance, and resource use efficiency. The plant is not a machine with independent parts; it is a living system whose power lies in the seamless coordination of its many processes.

Key Integration Points

- Light → Photosynthesis → Sugars
- Sugars → Respiration → ATP
- ATP → Active Transport → Minerals
- Minerals + Sugars → Amino Acids
- Hormones → Regulate All Above
- Stress → Modifies All Above

"The plant is a unified system — perturb one process, and all others respond."

~30

ATP per Glucose

ATP molecules generated from complete aerobic respiration of one glucose molecule

17

Essential Elements

Mineral elements whose absence prevents completion of the plant life cycle

5

Major Hormones

Core phytohormone classes coordinating growth, development, and stress

~1%

Water Use Efficiency

Only ~1% of absorbed water is used in photosynthesis; rest lost via transpiration